

Conflux Collective: A Student-Driven Model for HCI Education Through the Lens of Art-Tech

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Figure 1: A collage showing various Conflux Collective activities. Top row (left to right): Dall-E workshop, "Visual Perception & Art" workshop with Seth Riskin, *Movement, Sound, and Water in the Cage* prototyping, *Feed the Trees* design process; middle row (left to right): *Borealis* field trip, members of *Musical Chairs* project team, *Apocalypse* event, *Growing Your Tree* exhibition view, *Wormholes* installation, *#showerthoughts* user testing; Bottom row (left to right): *Passage to Possibility* installation, collaborative learning, *Inner Piece* exhibition view, *Touch Grass* demo, projection mapping workshop with Devon Bryant.

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3706666>

Abstract

As emerging technologies disrupt traditional ways of innovating and creating, higher education will require new models of interdisciplinary learning. In this case study, we review the operating model of Conflux Collective, a student-driven organization at Harvard University that facilitates interdisciplinary human-computer interaction (HCI) education through art-tech projects. We then outline our organizational impact, challenges, and strategic recommendations for designing HCI-inspired organizations. Our findings aim to guide educators, student organizers, and institutions interested in developing similar programs that embrace disciplinary diversity.

CCS Concepts

• **Human-centered computing** → **Participatory design**; *User centered design*; *Activity centered design*; **Interaction design theory, concepts and paradigms**; **Interaction paradigms**; **HCI theory, concepts and models**; *HCI design and evaluation methods*; • **Social and professional topics** → **Informal education**; • **Applied computing** → **Arts and humanities**.

Keywords

higher education, arts and humanities, creative technology

ACM Reference Format:

Peggy Yin, Alice Cai, Sofia Chen, Aida Baradari, and Xinyi Gao. 2025. Conflux Collective: A Student-Driven Model for HCI Education Through the Lens of Art-Tech. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3706599.3706666>

1 Introduction

Critical thinking, creativity, collaboration, communication, and technological literacy have been named as the most important 21st-century skills [19]. Best acquired through interdisciplinary curricula that focus on higher-order thinking and increased cultural awareness and competence, such skills help students navigate contemporary work environments, which demand a comfort with uncertainty and a willingness to cross traditional disciplinary boundaries [13, 33]. Despite this, higher education has remained structured around disciplinary silos [15, 16, 18]. In 2022, we—students at Harvard University—founded Conflux Collective to provide an interdisciplinary co-learning space supporting innovative human-computer interaction (HCI) projects by which students could hone their critical thinking, creativity, collaboration, communication, and technological literacy skills [9]. In this case study, we present opportunities and challenges for such student-driven educational models, and open-source our institutional knowledge.

2 Conflux Collective

Conflux Collective is a student-driven group of artists, designers, and engineers based in Cambridge, MA that creates at the intersection of arts and technology, with mediums that span sound art, data visualization, AI, XR, interactive art, performance, film, and more. The name, Conflux, is short for confluence—defined as ‘the act or process of merging.’ It references the participatory values embodied by the Fluxus artists of the 1960/70s and similarly emphasizes co-creation through experimentation (con- ‘together’ + fluxus).

Conflux was founded against a backdrop of technological advances democratizing access to engineering education. The release of generative AI models such as OpenAI’s GPT-3 and DALL-E 2 allowed nontechnical users to create complex visuals, coherent texts, functional code, and musical sound clips with unprecedented ease, lowering the entry barrier for a wide range of creative and technical activities [10, 14]. Internet access to information empowered self-directed research across diverse domains, while open source movements and advancements in hardware and software development tools enabled easier access to tools for prototyping (e.g., ever-cheaper 3D printers), and expanded opportunities for co-learning,

co-creating, and collaboration [28]. Yet many students were not aware or actively experimenting with such technologies. The founding goals of Conflux Collective were thus to:

- (1) Build a **community** for students to meet collaborators from diverse disciplinary backgrounds
- (2) Increase on-campus **awareness** of interdisciplinary ways of thinking, knowing, building, and being
- (3) Organize a resource **ecosystem** of equipment, advisors, and funding to support interdisciplinary activities
- (4) Launch **opportunities** for students to collaboratively develop novel, interdisciplinary work in HCI and art-tech
- (5) Train the **next generation** of student leaders who will lead these efforts at their institutions and beyond

3 Methods

Conflux contributes to the emerging phenomenological paradigm of HCI that forefronts human meaning and experience in the development of technological systems [23]. Structurally, Conflux organizes cross-institutional infrastructures, educational workshops, team projects, community activities, group exhibitions, residency programs, and learning opportunities for students to engage experientially with perspectives and methods across the arts and engineering sciences (Appendix A). Dynamic experimentation, iterative design, and rapid prototyping across a range of media and tools push students to develop core competencies in critical thinking, creativity, collaboration, communication, and technological literacy, while the Conflux culture encourages interactions between different ways of thinking and building, helping students embrace the ambiguity that comes with working in an emerging field such as art-tech. In this section, we describe the art-tech project structure (3.1), the rationale behind the types of activities that we host (3.2), the higher-level operations of our organization (3.3), and what distinguishes Conflux from other HCI and art-tech programs (3.4).

3.1 Anatomy of an Art-Tech Project

Art-tech projects can vary widely in team structure, development processes, and output (Fig 2). Projects are usually structured around Technical, Creative, and Cross-functional needs (e.g., recruitment, resource allocation, parts integration, etc.). Once conceptualized, projects follow an HCI prototyping process: ideation, design, prototyping and user testing, and then rapidly iterating between design and prototyping until satisfactory performance is reached (Fig 3).

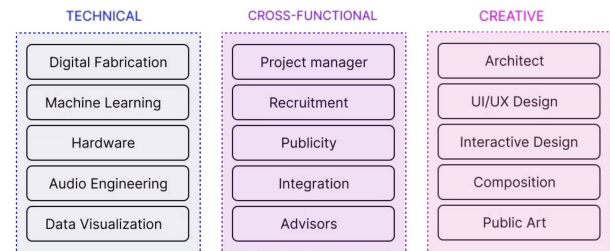


Figure 2: An example of a team diagram from project *Musical Chairs*. Students can occupy multiple roles across subteams.

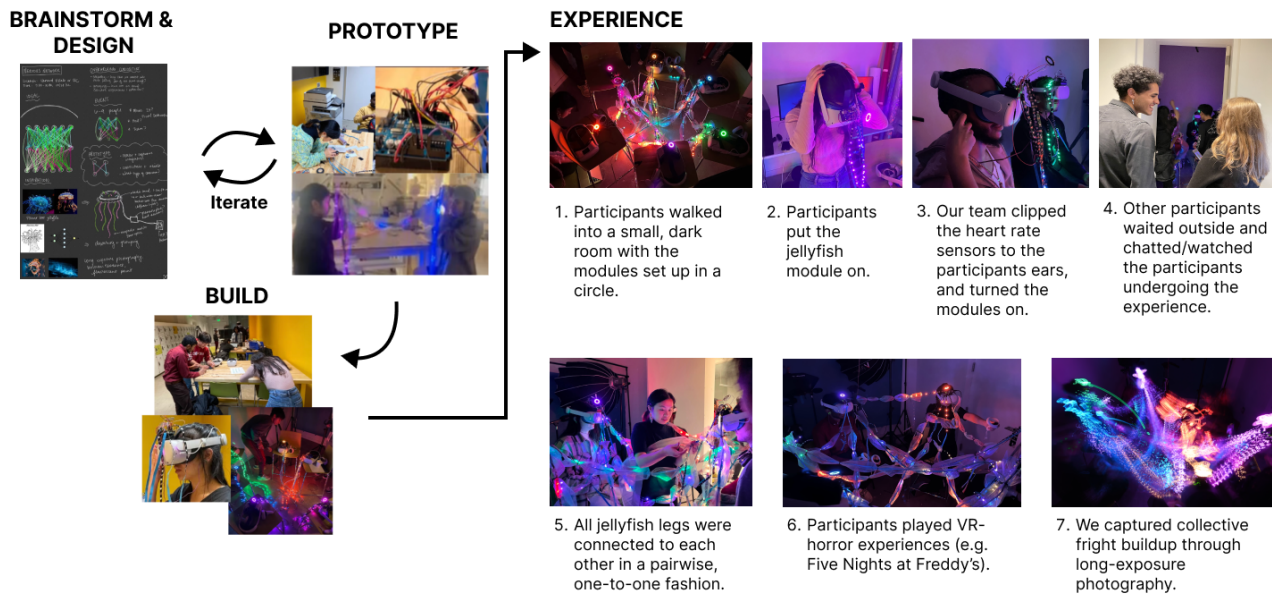


Figure 3: Example community event demonstrating the anatomy of an art-tech experience. *Nervous Network* was a Halloween-themed event in which people played horror games in VR while hooked up to biosensors and connected to each other to form a web of heartbeat-mirroring lights. Groups collectively created art reflecting their fright buildup, which was then captured by long-exposure photography. While waiting to participate, others at the event could watch the light show. A pre-event workshop engaged students in the processes of prototyping and interaction design as students built the wearables together.

3.2 Student Activities

- **Events:** Events, modeled after Fluxus happenings, focus on showcasing prototypes of interactive media or creative technology [32]. For example, *Notes on Love* collected and displayed student-submitted love letters through live data visualizations. Events are well-suited for observing user interactions in natural settings and support student evaluations of their designs in a low-stress, communal environment.
- **Workshops, Field Trips, and Studio Visits:** Hands-on workshops introduce students to art-tech skills in co-learning environments. The *Dall-T* workshop introduced participants to AI tools and dye sublimation through a personalized t-shirt design project. Field trips, such as the visit to the *Brain, Body + Breath* multisensory concert at the MIT Museum, broaden exposure to professional creative processes and finished projects. These experiences also deepen students' understanding of the many media encompassed in art tech.
- **Learning Groups:** Open to all experience levels and backgrounds, these semester-long groups emphasize experiential learning and interdisciplinary exploration. Co-designed and directed by students from cross-disciplinary backgrounds, the groups infuse artistic and technical methods in their explorations of a central topic or medium. For example, the *Dream Team* learning group studied the connection between dreams and creativity through weekly discussions, experimental demos, and experiential events like "dreamovers," where members practiced dreaming techniques together.

- **Residency Program:** The residency provides an immersive and intimate art-tech learning experience for a committed cohort of student residency fellows (Appendix B). Residency fellows live on campus, engaging in programming centered around a theme exigent to arts and technology, such as the 2024 *Symbiotic Perceptions* residency exploring tech's intersection with more-than-human and posthuman philosophies (Appendix C). The first week of residency emphasizes thematic learning through trips, lectures, and workshops with experts, while the second week focuses on project development alongside mentors and guest critics. Fellows showcase in a culminating group exhibition, produced in collaboration with the residency directors, the residency fellows, and production-oriented Conflux members.
- **Yearlong Projects:** These intensive projects bridge theory and practice in collaborative settings, and involve conceptualizing, designing, and installing works in public or professional spaces like museums. Teams receive resources, funding, and mentorship to explore cutting-edge technologies, and they work together to complete a project within a specified production timeline. Project outputs range from interactive new media installations to creative technology tooling to public art. Project participants gain skills in research, self-directed learning, technological development, leadership, user testing, and project management. Project teams also have an opportunity to observe how the public interacts with their work.

3.3 Structure & Operations

Extracurricular organizations serve as flexible containers for interdisciplinary, project-based, informal learning, offering students opportunities for leadership and collaborative problem-solving skills development [11, 25, 27, 30]. Such learning opportunities may be especially attractive to younger generations, who tend to favor active, hands-on, peer-to-peer learning over traditional instruction [31]. Conflux is structured as an extracurricular organization for students to engage in peer-to-peer informal learning through collaborative, student-directed projects. The organizational structure also presents an opportunity for bridging the arts and engineering sciences, which are often siloed by departments housed on opposite ends of campuses, curricular barriers preventing access to resources outside of one's chosen field (e.g., archives, makerspaces, etc.), and social groups clustering around disciplines [24].

The Conflux board functions as a central governing body enabling interdisciplinary student activities through fundraising, grant writing, budgeting, materials, publicity, partnerships, and general administrative overhead (Appendix B). Executive board members are in charge of the organization's day-to-day operations, while an advisory board of former board members and university faculty and staff members from HCI, computer science, arts, and engineering departments provide guidance. Notably, most board members are also actively involved as participants or leaders in Conflux activities, facilitating board decision-making.

3.3.1 Funding. Conflux acquires funding primarily through university grants for student organizations, including semesterly funding from the Harvard Undergraduate Association, the Paulson School of Engineering and Applied Sciences, the Harvard Office for the Arts, as well as one-off grants and individual donations. The semester budget is finalized 1.5 months into each semester.

3.3.2 Selection Processes. Applications are only required for activities that are limited by funding or resource constraints (e.g., residency). Applicants are evaluated according to criteria of expressed curiosity, creativity, team orientation, the potential impact of the program on their lives, and the unique perspective they might contribute to the activity group.

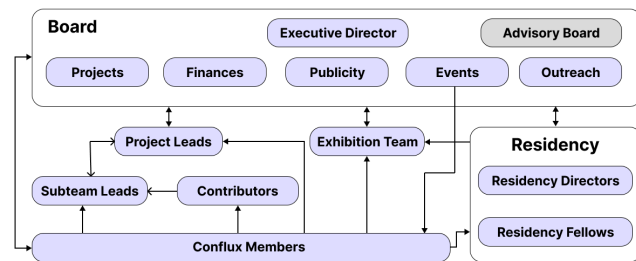


Figure 4: Flowchart of Conflux's organizational structure. Most members of the board also lead Conflux activities, and members can assume non-board leadership positions such as Project Leads or Residency Directors. The executive board is accessible to all members, while the advisory board engages on an as-needed basis with the executive board.

3.4 Comparative Analysis

Art-tech programs have historically embraced different definitions, approaches, and models. For example, the Experiments in Art and Technology program at Bell Labs and the Art & Technology program at Los Angeles County Museum of Art brought professional artists and engineers together to apply new technologies towards creative expression, resulting in new media sculptures, installations, and performances [12, 26]. HCI research laboratories such as the MIT Media Lab publish papers that document design methods for tools that augment creative expression [1, 4, 5]. Formal learning programs such as the New York University Interactive Telecommunications Program / Interactive Media Arts Program provide "an art school for engineers" and "an engineering school for artists" to separately train graduate and undergraduate students to develop of creative interactive systems [2, 3]. In contrast, Conflux activities are forms of informal learning primarily serving undergraduate students, with graduate students and professionals participating alongside undergraduates. Conflux also deliberately maintains an open-ended maker culture that does not restrict students to specific outputs, leading to final projects that span output formats typically seen from many different art-tech models.

4 Results

Since Conflux's founding, we have hosted 5 projects, installed 6 times in public. Outside of residency, we have run 12 workshops, 3 field trips, and 5 events, including 2 aimed at addressing issues of diversity, equity, inclusion, and belonging in the art-tech field at large. Over 2 residencies, we have hosted 32 students and incubated 10 team projects. We have produced 6 community-sourced projects, and hosted 2 learning groups engaging 14 students each. 75 students representing over 20 different majors have participated in projects showcased at venues such as Harvard's Arts First festival, the MIT Museum's After Dark series, Harvard's Design and Project Fair, Cambridge Science Festival, as well as our 2 own exhibitions at the Paulson School of Engineering and Applied Sciences and the Student Organization Center at Hilles (Liminal Interfaces and Symbiotic Perceptions, respectively) [6–8]. Conflux has been featured in the *Paulson School of Engineering and Applied Sciences Newsroom*, *The Crimson*, and *C-Print Journal* [17, 20–22, 35]. Conflux leadership has been invited to present the organizational model and projects at the Distinguished Presenter Series at the Harvard Graduate School of Education as well as the Salata Institute Spring 2024 Student Event [29].

4.1 Design Space for Student Engagement

Over time, we observed that engagement with each of these activities tended to differ based on required time commitment and what students were looking for. We distilled various engagement styles into five archetypal "modes":

- (1) Exploration: students test and experiment with the variety of processes and media used in HCI and art-tech
- (2) Skill-building: students learn specific HCI and art-tech skills through hands on activities
- (3) Collaboration: students work on projects with students from other disciplines

- (4) Production: students direct or produce interdisciplinary programming or exhibitions
- (5) Board: students drive the vision, mission, and growth of the organization at large

For those looking to find the best activities for their art-tech communities, we synthesize Conflux activity attributes and how they address specific community needs, student modes of engagement, and leadership capacity in Figure 5.

4.2 Stakeholder Feedback

To better understand Conflux's impact, we compiled and analyzed interviews, feedback, and testimonials collected from Conflux members and institutional stakeholders who have participated in, supported, and funded Conflux activities over our two years of operation.

4.2.1 Participants. Students engaged in Conflux primarily because they were in search of a like-minded community. As one residency fellow remarked, "the opportunity to get together with a group of people from many different backgrounds and collectively push the boundaries of what we can create, what we should create, and what is meaningful to create... The openness of ideas and the shared desire to really do something new and interesting was a really exciting environment to be in." Conflux events have facilitated casual interactions among students from many disciplines and often served as the entry point for students to engage further with HCI or art-tech methods and tools. For example, many students leaving our annual Dall-T workshop remarked having their "minds blown" by the quality of AI art generators (especially the first time we ran it in 2022) and subsequently using them in their own time. One newcomer remarked: "I'm interested in making things because of Conflux. It made me realize that there are a lot of things that I can do if I just try to learn them."

Students from artistic backgrounds reported that Conflux programming exposed them to new ways of thinking about their artistic practice, stating: "Conflux helped me understand that what I love about creating music isn't necessarily the music itself. It is the immersiveness of crafting and entering a new space." Conflux programming also helped students from engineering backgrounds apply familiar skills or technologies in ways that they had not considered before. One residency fellow remarked: "Conflux was something I didn't know I needed...I realized I didn't actually really care about studying pure computer science and that there was something fundamentally misaligned about the classical course structure and concentration structure for CS...I had always been more interested in the design side of things in the CS concentration." Many students echoed this sentiment, expressing that they did not feel like they solely identified as an engineer or as an artist, and appreciated the Conflux culture as one in which they could explore their full spectrum of interdisciplinary interests. Often, students recalled their time in Conflux as profoundly transformational, citing the programming as critical to their college major or career decisions.

Student leaders also found their role in facilitating this transformation rewarding. As one residency director observed: "Our organization has a impact on their choice of major and even life path. Some of the residents want to change their major because of

the residency, and some of them decide to take on a second major or path of exploration related to creative practice. That is very important for us."

4.2.2 Alumni. Conflux alumni frame the organization as their focused creative outlet during college. Some mentioned that Conflux's focus on design before development helped them develop better creative problem skills that they could apply on the job, with many mentioning their Conflux projects during job interviews. Moreover, alumni agreed that Conflux expanded their conceptions of what a career could include, commenting that Conflux "[broadens] the possibility space for what a technologist can look like."

4.2.3 Institutional Supporters. Conflux activities were supported by a variety of institutional stakeholders spanning departments, offices, and administrators in the arts and engineering sciences. Institutional supporters found our commitment to disciplinary diversity particularly compelling, and emphasized that it aligned with their goals to expand campus opportunities for students to meet other students from different backgrounds. Other administrative offices emphasized Conflux's technical focus on building, or artistic focus on "what technology can teach us about the nature of creative thought" [34]. Institutional stakeholders also commented on their support for projects like PHOTOSYNTHESIS, which test solutions to real-world problems such as climate change through the lens of the innovative arts.

5 Challenges & Opportunities

We encountered the following key challenges:

- **Incentives Alignment:** Our diverse membership shared an interest in art-tech as a field, but it took iterative experiments to clarify how different stakeholders' incentives could be satisfied. Specifically, while we internally sought to foster a culture that rewarded interdisciplinarity, the external systems we were bridging maintained divergent priorities. Engineering ecosystems often emphasized working on technical challenges, while creative ecosystems valued meaningful creative experiences. Academic research ecosystems pushed for peer-reviewed publications, whereas arts ecosystems pushed for exhibitions or shows. These divides complicated decisions on our organizational outputs and definitions of success.
- **Process Differences:** Our emphasis on cross-functional teams often led to methodological frictions. These issues were most apparent during ideation, where students' varied approaches to generating ideas came into conflict. For instance, during an ideation session, a student with a choreography background expressed discomfort with using the disembodied techniques of rapid prototyping on a whiteboard. Terminological differences also surfaced, especially on projects requiring niche expertise. In response, team leaders were encouraged to advise students to expect and lean into these frictions as possibilities for further creativity and learning. In response to student feedback, we also adjusted our recruitment process for yearlong projects to include multiple entry points, so that students could choose to engage only in the parts of the project that interested them most (e.g.,

Activity	Commitment	Involvement	Purpose	Examples (Appendix A)
Events \$50 – \$250 <i>One-night gatherings to engage the art-tech community-at-large</i>	Organizers Medium Participants Low	When: Semester Lead Time: 1 week Community: all	Modes: Explore, Skill-build, Collaborate Audience: Public Learn: how to design and test prototypes for live settings	- Nervous Network - Notes on Love - Sisters with Transistors - Apocalypse
Workshops \$0 – \$100 <i>Workshops, field trips, and studio visits to learn from experts</i>	Organizers Low Participants Low	When: Semester Lead Time: 2 weeks Community: students	Modes: Explore, Skill-build Audience: Public Learn: how experts are contributing to the field at large	- Dall-T - Mechanical Flowers - Brain, Body + Breath - Neurafutures - Borealis
Projects \$500 – \$1500 <i>Collaborative projects with professional showcase opportunities</i>	Organizers High Participants Low, Medium, High	When: Semester Lead Time: Variable Community: project teams and leads (application-based)	Modes: Skill-build, Collaborate, Production Audience: students with relevant expertise and interest Takeaways: how to go from idea to design, iteration, and implementation	- Musical Chairs - Wormholes - PHOTOXSYNTHESIS - Black Box Gallery - Psychosymphonies
Learning Groups ~\$1000 <i>Semesterly groups, learning and brainstorming around an interdisciplinary topic</i>	Organizers Medium Participants Low	When: Semester Lead time: Variable Community: students	Modes: Explore, Skill-build, Production Audience: students Takeaways: how to engage in interdisciplinary thinking and ideation	- Dream Team - Affective Robotics
Residency + Group Exhibition ~\$4000 <i>Two-week long educational programs to incubate interdisciplinary projects</i>	Organizers High Participants High	When: Winter Break Lead time: 2 months Community: residency fellows and directors (application-based)	Modes: Skill-build, Collaborate, Production Learn: how to build and engage with art tech from both theorists and practitioners	- Liminal Interfaces (Residency 2023) - Symbiotic Perceptions (Residency 2024)

Figure 5: A chart describing Conflux activities, including cost, time commitment, involvement, event purpose, and examples. The involvement column highlights the expected timeline and community stakeholder engagement for each of the activities. The last column provides Conflux examples, expanded on in Appendix A.

conceptualization or build), rather than having to commit for the full project timeline.

- **Resource Challenges:** Our projects required working with various institutional partners that were otherwise disconnected. For example, a sound art project might require audio equipment from the library media reserves, and audio wires from the electrical engineering laboratories; a robotic sculpture project might require consultation from graduate students and professors in robotics, and artists working in the plastic arts. Initially, we had a "tech director" to help locate resources, but later found project leaders better suited for this role due to their familiarity with project needs. Past project leads now advise new projects to streamline resource and mentorship searches. We recommend collaborating with administrators to map campus resources for easier access to materials and expertise.
- **Establishing & Transitioning Leadership:** Formalizing our board was challenging. The founding board in Fall 2022 consisted of friends who shared a common interest in art-tech and interested students recruited on a casual basis, leading to confusion over roles as administrative needs grew. In Spring and Fall 2023, we recruited a formal board, but struggled to define whether board members should lead activities or focus on administration. By Spring 2024, we clarified that board roles should center on administration, so non-board roles (e.g., project leads) could focus on production. Furthermore, extending the leadership transition over a longer

period of time allowed for a smoother transition, as we were able to shuffle roles and responsibilities until we reached the role clarity needed to efficiently support the ongoing student activities. We recommend future organizers experiment with transition timings, and to account for an onboarding timeline of at least half a semester.

- **Measuring Impact:** We developed metrics along three axes: project outputs, personal growth, and community impact. Sometimes these metrics conflicted, such as when deciding whether residencies should focus on production (emphasizing team outputs) or education (focusing on learning experiences). We used surveys, interviews, and demographic data to assess impact. For example, for residencies and projects, we collected information on student majors, class year, and previous technical and artistic activities, and we solicited qualitative feedback throughout our programming. However, more objective metrics could strengthen our approach.

5.1 Design Recommendations

For organizers seeking to establish student HCI / art-tech organizations, we recommend the following strategies:

- **To spark interest in art-tech in your community, host a workshop or event.** These can run for minimal or no funding (depending on institutional equipment and space access), with a dedicated team organizing 2-4 weeks in advance. Themes around popular emerging technologies (like newly released AI models, XR devices, etc.) and current events (like

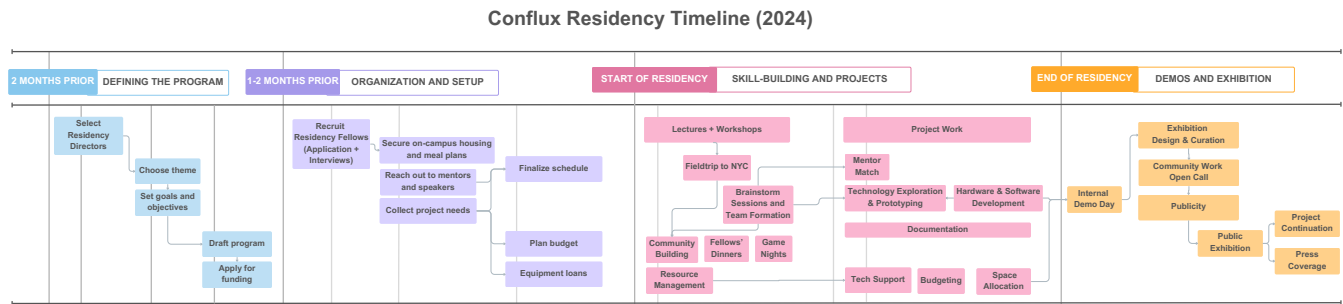


Figure 6: A timeline describing the residency planning and implementation process.

holidays) could attract students. For example, DALL-T leveraged the rising interest in OpenAI's models; *Nervous Network* leveraged the holiday spirit of Halloween.

- **To build a close community over a short period of time, host a residency.** Residencies can be advertised as a career opportunity for students seeking winter or summer break positions. Residency directors should start preparing 2+ months in advance, setting a thought-provoking theme, securing housing, food, and equipment for the program (e.g. university accommodations, meal plan access), seeking mentors, publicizing applications, interviewing and selecting applicants, and drafting programming (see Fig. 6 for an example timeline). We recommend hosting residencies during times that do not clash with academic calendars (e.g., during school breaks), to enable residency fellows and directors to focus on the residency experience and projects.
- **To generate specific outputs (e.g. papers, technologies, exhibitions), run a project or project-based organization.** Projects are generally initiated by 1-2 leaders, who define the vision, outputs, and timeline (typically before the semester starts or at the beginning of semester). Project leaders can run their own open-ended or referral-based recruiting process, and structure their team and work process how they see fit. Some larger projects benefit from multiple sub-teams, while others benefit from a single tight-knit and agile team. Recruiting project co-leads from different backgrounds has helped bridge knowledge gaps and fostered improved team communication and management.
- **To focus on the process and educational value of art-tech explorations, host a learning group.** Learning groups have the lowest administrative overhead. A couple people interested in the same topic can co-initiate a group and send an open invite to the student body. Examples include discussion-based groups—in which students read, watch, listen to, and discuss media of interest—and activity-based groups, in which students share activities, projects, and self-experiments with each other. *Dream Team*, for example, mixed these two modalities into a weekly discussion and activity group where participants discussed dream science papers and experimented with lucid dreaming techniques.
- **To start a sustainable organization in your community, evaluate your capacity to organize and your community members' interest in the types of activities listed**

in Figure 5. While we recommend board roles and responsibilities in Appendix B, we emphasize that the makeup of the executive board should be largely determined by the desired engagement modes of both board members and community members. Key to the sustainability of an organization is also institutional support. This support is best established through building relationships with relevant administrative departments and interested faculty advisors.

- **To publicize programming, try a variety of styles and framings that capture diverse audiences across disciplines.** To source organizers or board members, watch for highly engaged participants as potential leaders. To source funding, check for available university grants or corporate sponsorships. When funding is difficult to source, implement low-cost activities such as workshops, learning groups, and digital projects.
- **To evaluate your group's impact over time, track demographic data and engagement trends.** We also recommend organizers think explicitly and early on about what kinds of impact they aim to promote, especially if conflicting objectives present the possibility for later tension.
- **To expand your group's reach, host public events with opportunities for participation.** Our public events engaged a wide array of community members, including local residents, working professionals, and families. The addition of an "open projector" (akin to an open mic) to events like *Apocalypse* allowed community members at large to present their work alongside student projects, and open calls for community projects provided opportunities for community members to see their work produced and showcased in a public exhibition. Thus, events are an effective method for communities beyond institutional settings to connect with students in HCI, creative technology, and art-tech.

6 Limitations

The clash of diverse and often opposing worldviews, methods, definitions of success, and incentives within teams often led to deeper insights or new understandings between team members. However, dissatisfaction or mismatch between participants' expectations and experiences occasionally required compromise, especially regarding outputs. To address these tensions, we recommend implementing rapid feedback cycles, anonymous post-mortem surveys, and established mediator roles to foster a culture that embraces

methodological diversity. All in all, we believe tensions present an opportunity for organizations such as Conflux to continue iterating, updating, and co-designing cross-functional organizational processes to best support the investigation of generative frictions between disciplines.

We are grateful for the opportunity to report our learnings to date, and acknowledge that the speed and stability with which Conflux grew is owed in part to the institutional support and resources we received—including funding, spaces, and personal support from administrators keen on promoting experiential and interdisciplinary student experiences. Not all collegiate institutions may have this kind of support available, and certain components of our model may be more difficult to implement in less well-resourced environments. In the absence of university support, other private or philanthropic sources can potentially be looked towards. For administrator stakeholders, this limitation is a call for increasing the funding and mentorship available for this kind of innovation in higher education.

7 Conclusion

Art-tech initiatives like Conflux are proven vessels for HCI education. Through Conflux, students have developed and investigated varieties of human-technology systems and experiences, learned new skills, broadened their artistic and technological processes, and deepened their understanding of how art-tech and HCI may be used to investigate and expand our understanding of each other and the world around us. In this case study, we examined the wide array of activities, modes of engagement, stakeholders, and values Conflux has implemented over the years, and recommended plans of action for organizers seeking to build similar programs.

Acknowledgments

We would like to express our sincere gratitude to the Conflux community, mentors, sponsors, board, and alumni for their invaluable support and contributions. Their guidance, belief, and dedication made this program possible, and we are deeply appreciative of their efforts.

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A Appendix A: Examples of Student Activities

• Events:

- **Apocalypse:** a Boston-wide, themed art-tech meetup designed by Conflux Collective, with attendees from the MIT Media Lab, metaLAB, CAMLab, Boston Tech Poetics, Harvard's Graduate School of Design, and local community members. Attendees brought demos of their work and could also share their work on an open projector.

- **Notes on Love:** featured art-tech projects, refreshments, and data visualizations of community-contributed love letters, anchoring conversations on the theme of love.
- **Sisters with Transistors:** featured a fireside chat with two electroacoustic composers and a screening of *Sisters with Transistors*, a documentary highlighting the pioneering women of electronic music who challenged gender norms within the music industry and expanded the possibilities of sonic creativity.
- **Nervous Network:** an experimental social developing sensor-visualization technologies to illuminate shared experiences of fear. Participants built wearable VR biosensors, creating a network that lit up with increasing heart rates as they watched horror films together.
- **Workshops, Field Trips, Studio Visits:**
 - **Dall-T:** an annual workshop introducing participants to generative AI image tooling and dye sublimation, methods that are changing the nature of printed graphics and production. Participants first explore prompt engineering using Midjourney and Dall-E, then dye sublimate their AI-generated artworks onto T-shirts, sweaters, canvases, metal plates, and other materials to take home.
 - **Mechanical Flowers:** as a part of PHOTOSYNTHESIS, Conflux ran a series of 8 workshops throughout the Spring about building mechanical flowers, including workshops on using recycled materials to make flowers and 3D-printing gears to add different degrees of movement to the flowers. This workshop was also presented at the Cambridge Science Festival, serving as an educational and accessible way for people to get involved with a more technical project.
 - **Brain, Body + Breath** multisensory concert at the MIT Museum (Fall 2022) allowed students to engage with premieres of compositions blending music and technology in a live setting.
 - **Neurafutures and Borealis:** students explored an exhibit using 3D-printing technologies to create replicas of BCIs from science fiction, coupled with a behind-the-scenes introduction to such *Neurafutures* by curator and creator Nataliya Kosmyna. Students then headed outside to see an ongoing light art installation that emulated the Aurora Borealis.
- **Projects:**
 - **PHOTOSYNTHESIS:** an interactive installation exploring the potential of agrivoltaics in urban settings. The project features a “garden” underneath four RV-sized solar panels, which power various mechanical and digital plants—including rotating reflective flowers located on the top of the solar panels and digitally projected flora created by generative AI. These plants coexist with shade-tolerant perennial plants, illuminating the potential for clean energy innovation to be integrated with cutting-edge technical development.
 - **Psychosymphonies:** a live, electroacoustic performance piece premiered in Paine Hall. The work explored the biosignals present in our brains and our breaths, and used EEG sensors to control a synthesizer.
 - **Musical Chairs:** a participatory art installation augmenting furniture to support conversation emotionally. The work featured three custom-fabricated chairs augmented with time-of-flight sensors, microphones, and transducers, with the core engine of the installation being a synthesizer for interactive soundscapes that fostered the feeling of being heard.
 - **Wormholes:** a collection of interactive, audio-visual installations distributed across campus that examined themes of transience, connection, and serendipity. The wormholes—bi-directional camera feeds—allowed those separated by space-time to see and interact with each other in realtime.
- **Learning Groups:**
 - **Dream Team:** a weekly learning group studying contemporary research on the relationship between dreams, creativity, and collective dream practices. Activities included demos of experimental stimuli, meetups with leading dream researchers, and “dreamover” events where members could practice dreaming techniques together. This group functioned as a more casual and social group, fostering a supportive environment for experimenting and brainstorming project ideas, bridging theory and practice.
 - **Affective Robotics:** a project-based learning group discussing and prototyping evocative forms of soft robotic movement. Each week, the cohort engaged in 2-hour hands-on workshops centered on learning foundational skills in soft robotics, and a 2-hour discussion group examining historical and artistic formulations of the body.
- **Residency Program:**
 - *Symbiotic Perceptions:* (see Appendix C)
 - *Liminal Interfaces:* the inaugural residency focused on incubating a series of interstitial artworks—art that would be genre-bending, genre-breaking, or simply the art of the in-between. Each piece explored the ways in which art and media technology could transform liminal spaces—the transitory, ambiguous, and often painful spaces between locations or states of being—into personal, playful, and meaningful places for agency and growth. Fellows were first challenged to make meaning out of seemingly meaningless, perfunctory elevator interactions. Residency fellows then worked in teams to develop interfaces that addressed liminal spaces of their own choosing.

B Appendix B: Example Board Roles and Responsibilities

- **Director:** create and implement the vision of the organization. Direct members of the board to ensure organizational operations and activities run smoothly. Identify new opportunities for the club to grow and thrive.
- **Finance:** Maintain the annual budget, organizational bank account, and organizational expenses for all organizational activities. Initiate and manage partnerships and sponsorships. Write grant applications and oversee project inventory.
- **Events:** Plan creative and hands-on public events that introduce campus and community members to art-tech. Ensure

that all events are inclusive, safe and aligned with the organization's mission and objectives. Promote a sense of community within the organization and integration into campus culture.

- **Projects:** Manage, lead and recruit for projects. Oversee project leaders to ensure projects are completed on schedule and within budget. Document and preserve records of projects.
- **Publicity:** Lead and manage the creation and distribution of all publicity materials for the organization. Implement a publicity schedule and strategy for organizational events and recruitment efforts. Ensure that all publicity efforts align with the organization's mission and objectives.
- **Outreach:** Lead communication with other campus organizations, departmental faculty and staff, and other art-tech communities. Work closely with the rest of the board to find needed mentors and experts for projects, events, and residency.

C Appendix C: Examples of Residency Programming and Projects

C.0.1 Phase 1: Thematic Content.

- Talk: "TeleSymbiosis: Towards Human-Nature Interaction" by JB Labrune: Although the benefits of interacting with nature are well-known, very few technological environments are inspired by human-nature interactions. Recently, sustainable design and more-than-human design have called attention to the importance of nature in design philosophies. This talk will discuss some recent works pioneering such directions.
- Tour: Media Lab + NANO Immersion Lab Tour: Focusing on novel sensing technology, human computer interfaces for connection and communication, and the future of XR interaction modalities, the tour will showcase interaction models for expressing the human experience through virtual media. The tour of the MIT Media Lab will center on the work of the Tangible Media Group. Then, we'll take a walk to the MIT NANO Immersion Lab, a motion capture and technology performance space used for art and science collaboration.
- Workshop: "Visual Perception & Art" by Seth Riskin: A conceptual and practical exploration of light as the "substrate" of the visual arts, this workshop focuses on shaping visual perception through the artistic manipulation of light. The workshop touches on the MIT tradition of using light as an interdisciplinary language, bridging art and science. Participants will engage in hands-on experiments demonstrating technologies serving artistic purposes and artistic, experiential methods contributing to research in areas such as vision neuroscience.

C.0.2 Phase 2: Examples of Residency Projects.

- *Movement, Sound, and Water in the Cage*: "Fluid Symphony" is an interplay between movement, sound, and space. It's coupled with a reflection on the human perception of being caged in by feelings of the impending climate catastrophe.

"Fluid Symphony" uses sounds of water that change depending on the movement of a person engaging with the space to express how humans still have agency in improving the conditions of our climate. This concept crystallized from the belief that despite the constraints we might perceive, we as humans have the innate ability to transcend our realities. Through the fusion of water, movement, and sound within the spatial confines of the exhibit, "Fluid Symphony" looks to awaken a sense of empowerment, inviting participants to recognize their role in reshaping their perception of our future.

- *Feed the Trees*: an immersive and interactive VR experience where we delve into the intricate relationship between trees and humans, questioning the balance of this connection. Inspired by the idea of reciprocity, we wondered what it would mean if trees took back from humans what we've taken from them. In that case, what would it mean to save a tree? Using Unity, a VR environment is crafted where the landscape is colored in reds and pinks, a reminder of the blood that once flowed. In this new world, participants witness firsthand what it means to "feed the trees." As blood bubbles rise from their hands, their health deteriorates until the inevitable end. The digital realm is merged seamlessly with the physical with the use of projections and sound, creating a multi-dimensional experience that blurs the boundaries between reality and imagination. As participants immerse themselves in the virtual world, giving their blood to the tree, they feel the vibrations of the tree's heartbeat strengthening. Meanwhile, spectators outside the virtual space are treated to a mesmerizing display as the participant's hand shoots out a network of red from the tree's heart to the branches above, each beat synchronized with their own heartbeat. The tree itself is crafted from humble materials such as wire, cardboard pipes, and crumpled paper, representing the lesser condition from our world to the VR world. Even its heart is created from simple foil construction paper. Through this immersive art piece, viewers are invited to reflect upon the intricate dance of give and take between humanity and nature. It serves as a powerful reminder of our interconnectedness with the natural world and urges us to consider the consequences of our actions on the delicate balance of our ecosystem.
- *Riverside Funhouse*: Inspired by the ability of abnormal stimuli, from the uncanny to the supernatural, to manipulate our empathy for humanlike forms, this collage of mirrors – literal and figurative, familiar and unfamiliar, analog and digital – distorts the visual cues that define our expectations of our mirror images. *Riverside Funhouse* inverts the passive experience of happening upon one's reflection in your environment, be it in a body of water or in the glass of a darkened storefront, the position we occupy in the scenery brought into consciousness, to transform the intentional experience of observing one's reflection, in a mirror on the wall or at a vanity table, into an awareness of the influence of visual perceptions the construction of the empathetic perceptions we approach the world with.